

Local Moran's I

Where are the clusters? Lung cancer in Pennsylvania

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Where we left off

Global Moran's I told us that lung cancer SIRs in Pennsylvania are spatially autocorrelated: similar values tend to sit next to each other. It did not tell us *where*. Today we break that single number into 67 pieces, one per county. Reading: **Rogerson & Yamada Ch. 4**.

```
library(sf)
library(dplyr)
library(ggplot2)
library(patchwork)
library(spdep)
library(SpatialEpi) # pennLC

theme_set(theme_void(base_size = 12))
```

Data and weights (recap)

Same SIRs and queen-contiguity weights as last time, in one chunk.

```
data(pennLC)
d <- pennLC$data

ref <- d |>
  group_by(race, gender, age) |>
  summarise(r = sum(cases) / sum(population), .groups = "drop") # reference rates

county <- d |>
  left_join(ref, by = c("race", "gender", "age")) |>
  group_by(county) |>
  summarise(O = sum(cases), E = sum(population * r), .groups = "drop") |>
  mutate(SIR = O / E)

pa <- st_as_sf(pennLC$spatial.polygon) |>
  st_set_crs(4326) |>
```

```

mutate(county = names(pennLC$spatial.polygon)) |>
st_transform(5070) |> # Albers
left_join(county, by = "county")

nb <- poly2nb(pa, queen = TRUE)
lw <- nb2listw(nb, style = "W") # row-standardized

sir_breaks <- c(0, 0.67, 0.83, 0.95, 1.05, 1.2, 1.5, Inf)
sir_fill <- scale_fill_brewer(palette = "RdBu", direction = -1, drop = FALSE, name = "SIR")

moran.test(pa$SIR, lw)$estimate[1] # global I

```

Moran I statistic
-0.01055968

```

ggplot(pa) +
  geom_sf(aes(fill = cut(SIR, sir_breaks)), colour = "white", linewidth = 0.2) +
  sir_fill

```

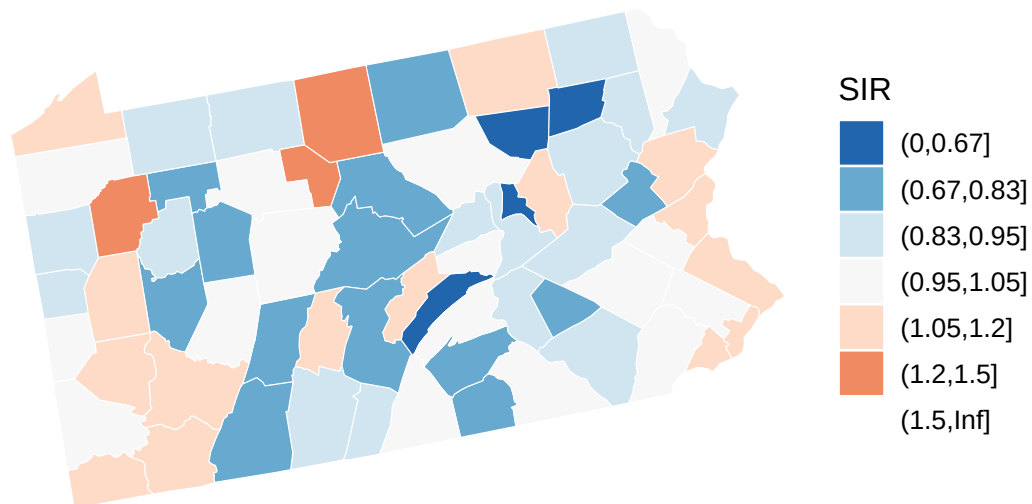


Figure 1: Lung cancer SIRs, Pennsylvania 2002.

From global to local

One number per county

Center the SIRs, $z_i = x_i - \bar{x}$. Global Moran's I (row-standardized weights) is

$$I = \frac{\sum_i z_i \sum_j w_{ij} z_j}{\sum_i z_i^2}.$$

The numerator is a sum over counties. Pull out each county's term and you get **local Moran's I** (Anselin 1995):

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j, \quad m_2 = \frac{\sum_i z_i^2}{n}.$$

In words: **(my deviation) × (my neighbours' average deviation)**, scaled by the variance. Averaging the I_i gives back the global I .

My SIR	Neighbours' SIR	Sign of I_i	Type
above mean	above mean	+	High-High
below mean	below mean	+	Low-Low
above mean	below mean	−	High-Low
below mean	above mean	−	Low-High

Computing it

`localmoran_perm()` returns I_i , a p -value, and the quadrant for every county.

```
set.seed(374)
lisa <- localmoran_perm(pa$SIR, lw, nsim = 999)

pa <- pa |>
  mutate(
    Ii      = lisa[, "Ii"],
    p_sim   = lisa[, "Pr(z != E(Ii)) Sim"],      # p-value
    quad    = as.character(attr(lisa, "quadr")$mean) # quadrant
  )
```

The LISA cluster map

Colour significant counties by quadrant; everything else grey.

```
lisa_cols <- c("High-High" = "#d7191c", "Low-Low" = "#2c7bb6",
              "High-Low" = "#fdae61", "Low-High" = "#abd9e9",
              "Not significant" = "grey92")

pa <- pa |>
  mutate(cluster = factor(if_else(p_sim < 0.05, quad, "Not significant"),
                          levels = names(lisa_cols)))
```

```

m_sir <- ggplot(pa) +
  geom_sf(aes(fill = cut(SIR, sir_breaks)), colour = "white", linewidth = 0.2) +
  sir_fill

m_lisa <- ggplot(pa) +
  geom_sf(aes(fill = cluster), colour = "white", linewidth = 0.2) +
  scale_fill_manual(values = lisa_cols, drop = FALSE, name = NULL)

m_sir | m_lisa

```

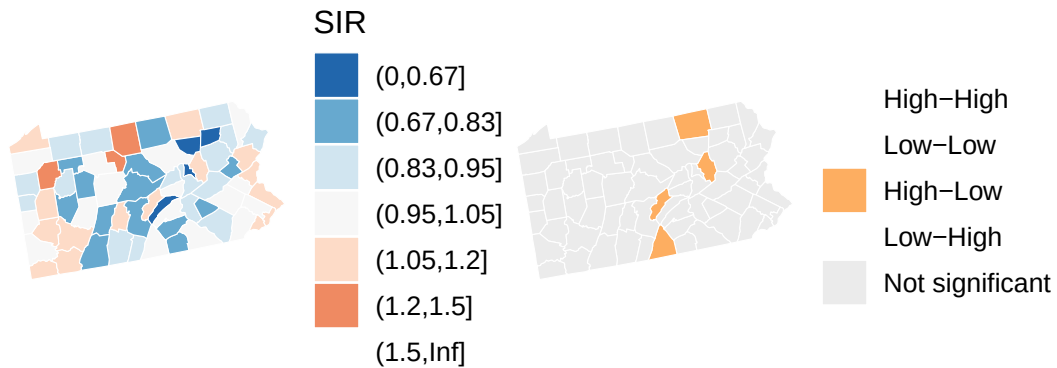


Figure 2: SIR (left) and LISA clusters at $p < 0.05$ (right).

```

pa |>
  st_drop_geometry() |>
  filter(cluster != "Not significant") |>
  arrange(cluster, p_sim) |>
  transmute(county, cluster, 0, E = round(E, 1),
            SIR = round(SIR, 2), Ii = round(Ii, 2), p_sim)

```

	county	cluster	0	E	SIR	Ii	p_sim
1	franklin	High-Low	105	109.9	0.96	-0.12	0.030
2	columbia	High-Low	58	53.3	1.09	-0.71	0.032
3	mifflin	High-Low	45	40.8	1.10	-0.95	0.042
4	bradford	High-Low	59	53.2	1.11	-0.89	0.048

Reading the map

- **High-High** (hot spot): high SIR, surrounded by high SIRs.
- **Low-Low** (cold spot): low SIR, surrounded by low SIRs.
- **High-Low / Low-High** (spatial outliers): a county that breaks from its surroundings.

Two cautions. A coloured county is the *core* of a local pattern; its neighbours are part of it even when they stay grey. And look at E_i in the table: a county with few expected cases can produce an extreme SIR, and a “cluster” by chance. We return to how many counties we are testing at once next session.

If you want to challenge yourself: build and read a LISA map

20 minutes.

1. Change the weights to the **four nearest neighbours**:

```
pts <- st_point_on_surface(st_geometry(pa))
nb4 <- knn2nb(knearneigh(st_coordinates(pts), k = 4))
lw4 <- nb2listw(nb4, style = "W")
```

Rerun `localmoran_perm()` with `lw4` (same seed) and redraw the LISA map. Which counties keep their label? Which change?

2. Pick one High-High county. Report its SIR, E_i , and the SIRs of its neighbours. Does the label make sense to you?
3. In one sentence for a county health officer, explain what a Low-High outlier means.

Key takeaways

1. Local Moran’s I splits global I into one value per county: own deviation $\times$ neighbours’ average deviation.
2. Positive I_i : like the neighbours (High-High, Low-Low). Negative I_i : unlike them (High-Low, Low-High).
3. The LISA map combines type and significance; read it alongside E_i and the choice of neighbours.

References

- Anselin, L. (1995). Local indicators of spatial association—LISA. *Geographical Analysis*, 27(2), 93–115.
- Rogerson, P., & Yamada, I. (2009). *Statistical detection and surveillance of geographic clusters*. Chapman & Hall/CRC, Ch. 4.